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Inventor(s)	YEH; Tefu et al.

METHOD OF FORMING NANOSTRUCTURE DEVICE AND RELATED STRUCTURE

Abstract

A method includes: forming a stack of alternating first semiconductor channels and second semiconductor layers on a substrate; releasing the first semiconductor channels by removing the second semiconductor layers; forming a gate dielectric on the first semiconductor channels; forming a transition metal nitride layer on the gate dielectric; and exposing the gate dielectric in a first region of the substrate by removing the transition metal nitride layer. The removing includes: performing a first etch using an acidic etchant; and after the performing a first etch, performing a second etch using an oxidizing etchant.

Inventors: **YEH; Tefu (Hsinchu, TW), CHENG; Chun-Chih (Hsinchu, TW), TU; Cheng-Chieh (Hsinchu, TW), HUNG; Jo-Chun (Hsinchu, TW), CHUANG; Ying-Liang (Hsinchu, TW)**

Applicant: **Taiwan Semiconductor Manufacturing Company Ltd.** (Hsinchu, TW)

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Background/Summary

BACKGROUND

[0001] The semiconductor integrated circuit (IC) industry has experienced exponential growth. Technological advances in IC materials and design have produced generations of ICs where each generation has smaller and more complex circuits than the previous generation. In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometry size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling down process generally provides benefits by increasing production efficiency and lowering associated costs. Such scaling down has also increased the complexity of processing and manufacturing ICs.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0003] FIGS. **1**A and **1**B are diagrammatic cross-sectional side views of a portion of an IC device according to embodiments of the present disclosure.

[0004] FIGS. **2**A-**13** are views of various embodiments of an IC device of at various stages of fabrication according to various aspects of the present disclosure.

[0005] FIG. **14** is a flowchart of a method of forming an IC device in accordance with various embodiments.

[0006] FIG. **15** is a flowchart of a method of forming an IC device in accordance with various embodiments.

DETAILED DESCRIPTION

[0007] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0008] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted

accordingly.
[0009] Terms indicative of relative degree, such as “about,” “substantially,” and the like, should be interpreted as one having ordinary skill in the art would in view of current technological norms.
[0010] The terms “first,” “second,” “third” and so on may be used herein to describe a sequence of events or sequential order of elements but may be exchanged or varied in some contexts. For example, a second layer may be formed on (e.g., sequentially after) a first layer, but in some contexts the first layer may be referred to as a “second layer,” “third layer,” “fourth layer” or the like, and the second layer may be referred to as a “first layer,” “third layer,” “fourth layer,” or the like.
[0011] The term “surrounds” may be used herein to describe a structure that fully or partially encloses another element or structure, for example, in three dimensions. For example, a first structure may “surround” a second structure on four lateral sides (e.g., left, right, front and back) without surrounding the second structure on two vertical sides (e.g., top and bottom). In other example, the first structure may wrap partially around the second structure, for example, by wrapping around three sides (e.g., top, front and back) while leaving other sides (e.g., left, right and bottom) exposed.
[0012] The present disclosure is generally related to semiconductor devices, and more particularly to field-effect transistors (FETs), such as planar FETs, three-dimensional fin FETs (FinFETs), or nanostructure FETs, such as nanosheet FETs (NSFETs), nanowire FETs (NWFETs), gate-all-around FETs (GAAFETs) and the like.
[0013] Dielectric layers, such as HfO2, SiO2 and ZrO2 offer benefits for turning on field effect transistors. Quality of the dielectric layers can include thickness, vacancies, heterogeneous atom contamination and the like, which can beneficial or detrimental effects on threshold voltage (Vt), channel resistance (Rch), ring performance (RO %), time-dependent-dielectric-breakdown (TDDB), maximum voltage degradation and the like of the FETs. Some of the detrimental effects can arise after ten years or more of operation of the FETs. In the case of nanostructure devices, such as nanosheet FETs, dielectrics at sheet corners undergo dry bombardment, which can result in a stress concentration effect and degradation of film quality. Then, a subsequent wet chemical etching or cleaning that is performed to remove metal can damage the underlying dielectric film. For example, N-type metal patterning can result in BARC (bottom anti-reflective coating) etching by plasma bombardment on dielectric layer, which can reduce film quality. In another example, metal etching by wet chemical(s) may have low selectivity to the dielectric layer due to reduced film quality.
[0014] Embodiments of the disclosure effectively reduce or eliminate damage to the dielectric layer by including a high-selectivity wet etching process that simultaneously etches metal without removing the dielectric layer. Nanosheet corners, which can be a hot spot for damage, represent a challenge for reliability testing. In the wet etching process, higher H.sub.2O.sub.2 concentration can slightly increase metal to dielectric film selectivity. In another example, slight acidic addition (e.g., HCl) can improve dielectric film quality by reducing damage thereto. The embodiments can provide benefits, such as SRAM (static random access memory) yield improvements due to reduced leakage current, improvements in dielectric quality, mobility increases and ring oscillator performance (e.g., RO %) increases, device lifetime increases, reduced voltage degradation and the like.
[0015] The nanostructure transistor structures may be patterned by any suitable method. For example, the structures may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the

patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the nanostructure structure.

[0016] FIGS. **1**A and **1**B are diagrammatic cross-sectional side views of a portion of a nanostructure device **10** in accordance with various embodiments. FIG. **1**A illustrates a view in an X-Z plane. FIG. **1**B illustrates a view in a Y-Z plane orthogonal to the X-Z plane. The nanostructure device **10** of FIGS. **1**A, **1**B is described in detail below to provide context for understanding the technical features and benefits of the various embodiments depicted in FIGS. **2**A-**13**. Source/drain region(s) may refer to a source or a drain, individually or collectively dependent upon the context.

[0017] Referring to FIG. **1**A, nanostructure devices **20**A, **20**B may be or include one or more N-type FETs (NFETs) or P-type FETs (PFETs). For example, the nanostructure device **20**A may be a PFET and the nanostructure device **20**B may be an NFET. The nanostructure devices **20**A, **20**B are formed over and/or in a substrate **110**, and generally include gate structures **200** straddling and/or wrapping around semiconductor channels **22**A, **22**B, **22**C, alternately referred to as "nanostructures," located over semiconductor fins **32** protruding from, and separated by, isolation structures **36** (see FIG. **1**B). The semiconductor channels **22**A, **22**B, **22**C may be referred to collectively as channels **22**. The gate structure **200** controls electrical current flow through the channels **22**A, **22**B, **22**C.

[0018] The nanostructure devices **20**A, **20**B are shown including three channels **22**A, **22**B, **22**C, which are laterally abutted by source/drain features **82**N, **82**P, and covered and surrounded by the gate structure **200**. Generally, the number of channels **22** is two or more, such as three or four or more. The gate structure **200** controls flow of electrical current through the channels **22**A, **22**B, **22**C to and from the source/drain features **82**N, **82**P based on voltages applied at the gate structure **200** and at the source/drain features **82**N, **82**P.

[0019] In some embodiments, the fin structure **32** includes silicon. In some embodiments, the nanostructure device **20**B includes an NFET, and the source/drain features **82**N thereof include silicon phosphorous (SiP), SiAs, SiSb, SiPAs, SiP: As: Sb, combinations thereof, or the like. In some embodiments, the nanostructure device **20**A includes a PFET, and the source/drain features **82**P thereof include silicon germanium (SiGe), either undoped or doped to form, for example, SiGe:B, SiGe:B:Ga, SiGe:Sn, SiGe:B:Sn, or another appropriate semiconductor material. Generally, the source/drain features **82**N, **82**P may include any combination of appropriate semiconductor material(s) and appropriate dopant(s).

[0020] The channels **22**A, **22**B, **22**C each include a semiconductive material, for example silicon or a silicon compound, such as silicon germanium, or the like. The channels **22**A, **22**B, **22**C are nanostructures (e.g., having sizes that are in a range of a few nanometers) and may also each have an elongated shape and extend in the X-direction. In some embodiments, the channels **22**A, **22**B, **22**C each have a nanowire (NW) shape, a nanosheet (NS) shape, a nanotube (NT) shape, or other suitable nanoscale shape. The cross-sectional profile of the channels **22**A, **22**B, **22**C may be rectangular, round, square, circular, elliptical, hexagonal, or combinations thereof.

[0021] In some embodiments, the lengths (e.g., measured in the X-direction) of the channels **22**A, **22**B, **22**C may be different from each other, for example due to tapering during a fin etching process (see FIGS. **3**A, **3**B). In some embodiments, length of the channel **22**C may be less than a length of the channel **22**B, which may be less than a length of channel **22**A. The channels **22**A, **22**B, **22**C each may not have uniform thickness (e.g., along the X-axis direction), for example due to a channel trimming process used to expand spacing (e.g., measured in the Z-axis direction) between the channels **22**A, **22**B, **22**C to increase gate structure fabrication process window. For example, a middle portion of each of the channels **22**A, **22**B, **22**C may be thinner than the two ends of each of the channels **22**A, **22**B, **22**C. Such shape may be collectively referred to as a "dog-bone" shape.

[0022] In some embodiments, the spacing between the channels **22**A, **22**B, **22**C (e.g., between the channel **22**B and the channel **22**A or the channel **22**C) is in a range between about 8 nanometers (nm) and about 12 nm, though ranges exceeding or below the said range may also be beneficial. In some embodiments, a thickness (e.g., measured in the Z-direction) of each of the channels **22**A, **22**B, **22**C is in a range between about 5 nm and about 8 nm, though ranges exceeding or below the said range may also be beneficial. In some embodiments, a width (e.g., measured in the Y-direction, shown in FIG. **1**E, orthogonal to the X-Z plane) of each of the channels **22**A, **22**B, **22**C is at least about 8 nm, however the width may be less than 8 nm in some embodiments.

[0023] The gate structure **200** is disposed over and between the channels **22**A, **22**B, **22**C, respectively. In some embodiments, the gate structure **200** is disposed over and between the channels **22**A, **22**B, **22**C, which are silicon channels for N-type devices or silicon germanium channels for P-type devices. In some embodiments, the gate structure **200** includes an interfacial layer (IL) **210**, one or more gate dielectric layers **600** on the interfacial layer **210** and a metal core layer **290** on the gate dielectric layer **600**. Additional layers, such as one or more work function tuning layers **900** (see FIG. **11**) may be present on the gate dielectric layer **600** between the gate dielectric layer **600** and the metal core layer **290**.

[0024] The interfacial layer **210**, which may be an oxide of the material of the channels **22**A, **22**B, **22**C, is formed on exposed areas of the channels **22**A, **22**B, **22**C and the top surface of the fin **32**. The interfacial layer **210** promotes adhesion of the gate dielectric layers **600** to the channels **22**A, **22**B, **22**C. In some embodiments, the interfacial layer **210** has thickness of about 5 Angstroms (A) to about 50 Angstroms (A). In some embodiments, the interfacial layer **210** has thickness of about 10 A. The interfacial layer **210** having thickness that is too thin may exhibit voids or insufficient adhesion properties. The interfacial layer **210** being too thick consumes gate fill window, which is related to threshold voltage tuning and resistance. In some embodiments, the interfacial layer **210** is doped with a dipole, such as lanthanum, for threshold voltage tuning.

[0025] In some embodiments, the gate dielectric layer **600** includes at least one high-k gate dielectric material, which may refer to dielectric materials having a high dielectric constant that is greater than a dielectric constant of silicon oxide (k≈3.9). Example high-k dielectric materials include HfO.sub.2, HfSiO, HfSiON, HfTaO, HfTiO, HfZrO, ZrO.sub.2, Ta.sub.2O.sub.5, or combinations thereof. In some embodiments, the gate dielectric layer **600** has thickness of about 5 A to about 100 A. The gate dielectric layer **600** may be a single layer or a multilayer.

[0026] The gate structure **200** also includes metal core layer **290**. The metal core layer **290** may include a conductive material such as Co, W, Ru, combinations thereof, or the like. In some embodiments, the metal core layer **290** is or includes a Co-, W- or Ru-based compound or alloy including one or more elements, such as Zr, Sn, Ag, Cu, Au, Al, Ca, Be, Mg, Rh, Na, Ir, W, Mo, Zn, Ni, K, Co, Cd, Ru, In, Os, Si, Ge, Mn, combinations thereof, or the like. Between the channels **22**A, **22**B, **22**C, the metal core layer **290** is circumferentially surrounded (in the cross-sectional view) by the one or more work function metal layers **900**, which are then circumferentially surrounded by the gate dielectric layers **600**, which are circumferentially surrounded by the interfacial layer **210**.

[0027] As depicted in FIG. **1**A, the nanostructure devices **20**A, **20**B may further include source/drain contacts **120** that are formed over the source/drain features **82**N, **82**P. The source/drain contacts **120** may include a core layer that is or includes a conductive material such as tungsten, ruthenium, cobalt, copper, titanium, titanium nitride, tantalum, tantalum nitride, iridium, molybdenum, nickel, aluminum, or combinations thereof. The core layer may be surrounded by one or more liner (or, “barrier”) layers, such as SiN or TiN, which help prevent or reduce diffusion of materials from and into the source/drain contacts **120**. In some embodiments, height of the source/drain contacts **120** may be in a range of about 1 nm to about 50 nm.

[0028] Silicide layers **118** may be positioned between the source/drain features **82**N, **82**P and the source/drain contacts **120**, at least to reduce the source/drain contact resistance. In some embodiments, the silicide layer **118** is or includes TiSi, CrSi, TaSi, MoSi, ZrSi, HfSi, ScSi, Ysi, HoSi, TbSI, GdSi, LuSi, DySi, ErSi, YbSi, or the like. In some embodiments, the silicide layer **118** is or includes NiSi, CoSi, MnSi, Wsi, FeSi, RhSi, PdSi, RuSi, PtSi, IrSi, OsSi, or the like. The silicide layer **118** may have thickness in a range of about 1 nm to about 10 nm. Thickness lower than about 1 nm may lead to an insufficient reduction in contact resistance. Thickness above about 10 nm may cause electrical shorting with the nanostructures **22**. In some embodiments, the silicide layer **118** is present below, and in contact with, etch stop layer **131**.

[0029] As depicted in FIG. **1**B, the nanostructure devices **20**A, **20**B may further include an interlayer dielectric (ILD) **130**. The ILD **130** provides electrical isolation between the various components of the nanostructure devices **20**A, **20**B discussed above, for example between neighboring pairs of the source/drain contacts **120**. An etch stop layer **131** may be formed prior to forming the ILD **130** and may be positioned laterally between the ILD **130** and the gate spacers **41** and vertically between the ILD **130** and the source/drain features **82**N, **82**P. In some embodiments, the etch stop layer **131** is or includes SiN, SiCN, SiC, SiOC, SiOCN, HfO2, ZrO2, ZrAlOx, HfAlOx, HfSiOx, Al2O3, or other suitable material. In some embodiments, thickness of the etch stop layer is in a range of about 1 nm to about 5 nm. In some embodiments, where the ILD **130** is not present (e.g., is removed completely prior to formation of the source/drain contacts **120**), the etch stop layer **131** may be in contact with the source/drain contact **120**. The etch stop layer **131** may be trimmed, for example, in the X-axis direction prior to formation of the source/drain contact **120** to improve fill quality of the source/drain contact **120**.

[0030] The nanostructure devices **20**A, **20**B include gate spacers **41** that are disposed on sidewalls of the metal core layer **290**, the gate dielectric layer **600** and the IL **210** above the channel **22**A, and inner spacers **74** that are disposed on sidewalls of the IL **210** and/or the gate dielectric layer **600** between the channels **22**A, **22**B, **22**C. The inner spacers **74** are also disposed between the channels **22**A, **22**B, **22**C. In the embodiment depicted in FIG. **1**A, the gate spacers **41** include a first spacer layer **41**A and a second spacer layer **41**B on the first spacer layer **41**A. The first and second spacer layers **41**A, **41**B may each include a dielectric material, for example a low-k material such as SiOCN, SiON, SiN, SiCN, SiOC or the like. In some embodiments, the second spacer layer **41**B is not present. Material of the first and second spacer layers **41**A, **41**B may be the same as or different from each other. In some embodiments, an upper portion of the second spacer layer **41**B (or the first spacer layer **41**A when the second spacer layer **41**B is not present) may be removed partially or fully to increase aspect ratio of an opening through which the source/drain region **82**N, **82**P is formed. FIG. **1**A depicts an embodiment in which the upper portion of the second spacer layer **41**B is not thinned.

[0031] FIGS. **14** and **15** depict flowcharts of methods **1000**, **2000** for forming an IC device or a portion thereof from a workpiece, according to one or more aspects of the present disclosure. Methods **1000**, **2000** are merely examples and not intended to limit the present disclosure to what is explicitly illustrated in methods **1000**, **2000**. Additional acts can be provided before, during and after the methods **1000**, **2000** and some acts described can be replaced, eliminated, or moved around for additional embodiments of the method. Not all acts are described herein in detail for reasons of simplicity. Methods **1000**, **2000** are described below in conjunction with fragmentary perspective and/or cross-sectional views of a workpiece, shown in FIGS. **2**A-**13**, at different stages of fabrication according to embodiments of methods **1000**, **2000**. For avoidance of doubt, throughout the figures, the X direction is perpendicular to the Y direction and the Z direction is perpendicular to both the X direction and the Y direction. It is noted that, because the workpiece may be fabricated into a semiconductor device, the workpiece may be referred to as the semiconductor device as the context requires.

[0032] FIGS. **2**A through **13** are views of intermediate stages in the manufacturing of FETs, such as nanostructure FETs, in accordance with some embodiments.

[0033] In FIG. **2**A and FIG. **2**B, a substrate **110** is provided. The substrate **110** may be a semiconductor substrate, such as a bulk semiconductor, or the like, which may be doped (e.g., with a p-type or an n-type dopant) or undoped. The semiconductor material of the substrate **110** may include silicon; germanium; a compound semiconductor including silicon carbide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor including silicon-germanium, gallium arsenide phosphide, aluminum indium arsenide, aluminum gallium arsenide, gallium indium arsenide, gallium indium phosphide, and/or gallium indium arsenide phosphide; or combinations thereof. Other substrates, such as single-layer, multi-layered, or gradient substrates may be used.

[0034] Further in FIG. **2**A and FIG. **2**B, a multi-layer stack **25** or “lattice” is formed over the substrate **110** of alternating layers of first semiconductor layers **21**A, **21**B (collectively referred to as first semiconductor layers **21**) and second semiconductor layers **23**. In some embodiments, the first semiconductor layers **21** may be formed of a first semiconductor material suitable for n-type nano-FETs, such as silicon, silicon carbide, or the like, and the second semiconductor layers **23** may be formed of a second semiconductor material suitable for p-type nano-FETs, such as silicon germanium or the like. Each of the layers **21**, **23** of the multi-layer stack **25** may be epitaxially grown using a process such as chemical vapor deposition (CVD), atomic layer deposition (ALD), vapor phase epitaxy (VPE), molecular beam epitaxy (MBE), or the like.

[0035] Three layers of each of the first semiconductor layers **21** and the second semiconductor layers **23** are illustrated. In some embodiments, the multi-layer stack **25** may include fewer or additional pairs of the first semiconductor layers **21** and the second semiconductor layers **23**. Although the multi-layer stack **25** is illustrated as including a second semiconductor layer **23** as the bottommost layer, in some embodiments, the bottommost layer of the multi-layer stack **25** may be a first semiconductor layer **21**.

[0036] Due to high etch selectivity between the first semiconductor materials and the second semiconductor materials, the second semiconductor layers **23** of the second semiconductor material may be removed without significantly removing the first semiconductor layers **21** of the first semiconductor material, thereby allowing the first semiconductor layers **21** to be patterned to form channel regions of nano-FETs. In some embodiments, the first semiconductor layers **21** are removed and the second semiconductor layers **23** are patterned to form channel regions. The high etch selectivity allows the first semiconductor layers **21** of the first semiconductor material to be removed without significantly removing the second semiconductor layers **23** of the second semiconductor material, thereby allowing the second semiconductor layers **23** to be patterned to form channel regions of nano-FETs.

[0037] In FIG. **3**A and FIG. **3**B, fins **32** are formed in the substrate **110** and nanostructures **22**, **24** are formed in the multi-layer stack **25** corresponding to act **1100** of FIG. **14**. In some embodiments, the nanostructures **22**, **24** and the fins **32** may be formed by etching trenches in the multi-layer stack **25** and the substrate **110**. The etching may be any acceptable etch process, such as a reactive ion etch (RIE), neutral beam etch (NBE), the like, or a combination thereof. The etching may be anisotropic. First nanostructures **22**A, **22**B, **22**C (also referred to as “channels **22**” below) are formed from the first semiconductor layers **21**, and second nanostructures **24** are formed from the second semiconductor layers **23**. Distance CD**1** between adjacent fins **32** and nanostructures **22**, **24** may be from about 18 nm to about 100 nm. A portion of the device **10** is illustrated in FIGS. **3**A and **3**B including two fins **32** for simplicity of illustration. The processes **1000**, **2000** illustrated in FIGS. **14**, **15** may be extended to any number of fins, and are not limited to the two fins **32** shown in FIGS. **3**A-**13**.

[0038] The fins **32** and the nanostructures **22**, **24** may be patterned by any suitable method. For example, one or more photolithography processes, including double-patterning or multi-patterning processes, may be used to form the fins **32** and the nanostructures **22**, **24**. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing for pitches smaller than what is otherwise obtainable using a single, direct photolithography process. As an example of one multi-patterning process, a sacrificial layer may be formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the fins **32**.

[0039] FIGS. **3**A and **3**B illustrate the fins **32** having tapered sidewalls, such that a width of each of the fins **32** and/or the nanostructures **22**, **24** continuously increases in a direction towards the substrate **110**. In such embodiments, each of the nanostructures **22**, **24** may have a different width and be trapezoidal in shape. In other embodiments, the sidewalls are substantially vertical (non-tapered), such that width of the fins **32** and the nanostructures **22**, **24** is substantially similar, and each of the nanostructures **22**, **24** is rectangular in shape.

[0040] In FIG. **3**A and **3**B, isolation regions, features or structures **36**, which may be shallow trench isolation (STI) regions, features or structures, are formed adjacent the fins **32**. The isolation regions **36** may be formed by depositing an insulation material over the substrate **110**, the fins **32**, and nanostructures **22**, **24**, and between adjacent fins **32** and nanostructures **22**, **24**. The insulation material may be an oxide, such as silicon oxide, a nitride, the like, or a combination thereof, and may be formed by high-density plasma CVD (HDP-CVD), flowable CVD (FCVD), the like, or a combination thereof. In some embodiments, a liner (not separately illustrated) may first be formed along surfaces of the substrate **110**, the fins **32**, and the nanostructures **22**, **24**. Thereafter, a core material, such as those discussed above may be formed over the liner.

[0041] The insulation material undergoes a removal process, such as a chemical mechanical polish (CMP), an etch-back process, combinations thereof, or the like, to remove excess insulation material over the nanostructures **22**, **24**. Top surfaces of the nanostructures **22**, **24** may be exposed and level with the insulation material after the removal process is complete.

[0042] The insulation material is then recessed to form the isolation regions **36**. After recessing, the nanostructures **22**, **24** and upper portions of the fins **32** may protrude from between neighboring isolation regions **36**. The isolation regions **36** may have top surfaces that are flat as illustrated, convex, concave, or a combination thereof. In some embodiments, the isolation regions **36** are recessed by an acceptable etching process, such as an oxide removal using, for example, dilute hydrofluoric acid (dHF), which is selective to the insulation material and leaves the fins **32** and the nanostructures **22**, **24** substantially unaltered.

[0043] FIGS. **2**A through **3**B illustrate one embodiment (e.g., etch last) of forming the fins **32** and the nanostructures **22**, **24**. In some embodiments, the fins **32** and/or the nanostructures **22**, **24** are epitaxially grown in trenches in a dielectric layer (e.g., etch first). The epitaxial structures may comprise the alternating semiconductor materials discussed above, such as the first semiconductor materials and the second semiconductor materials.

[0044] In FIG. **3**A and FIG. **3**B, appropriate wells (not separately illustrated) may be formed in the fins **32**, the nanostructures **22**, **24**, and/or the isolation regions **36**. Using masks, an n-type impurity implant may be performed in p-type regions of the substrate **110**, and a p-type impurity implant may be performed in n-type regions of the substrate **110**. Example n-type impurities may include phosphorus, arsenic, antimony, or the like. Example p-type impurities may include boron, boron fluoride, indium, or the like. An anneal may be performed after the implants to repair implant damage and to activate the p-type and/or n-type impurities. In some embodiments, in situ doping during epitaxial growth of the fins **32** and the nanostructures **22**, **24** may obviate separate implantations, although in situ and implantation doping may be used together.

[0045] In FIGS. **4**A-**4**C, dummy or sacrificial gate structures **40** are formed over the fins **32** and/or the nanostructures **22**, **24**, corresponding to act **1200** of FIG. **14**. A dummy or sacrificial gate layer **45** is formed over the fins **32** and/or the nanostructures **22**, **24**. The dummy gate layer **45** may be or include materials that have a high etching selectivity relative to the isolation regions **36**. The dummy gate layer **45** may be a conductive, semiconductive or non-conductive material and may be or include amorphous silicon, polycrystalline-silicond (polysilicon), poly-crystalline silicon-germanium (poly-SiGe), metallic nitrides, metallic silicides, metallic oxides, and metals. The dummy gate layer **45** may be deposited by physical vapor deposition (PVD), CVD, sputter deposition, or other techniques for depositing the selected material. A mask layer **47** is formed over the dummy gate layer **45**, and may include, for example, silicon nitride, silicon oxynitride, or the like. In some embodiments, a gate dielectric layer **43** is formed before the dummy gate layer **45** between the dummy gate layer **45** and the fins **32** and/or the nanostructures **22**, **24**. In some embodiments, the mask layer **47** includes a first mask layer **47**A in contact with the dummy gate layer **45**, and a second mask layer **47**B overlying and in contact with the first mask layer **47**A. The first mask layer **47**A may be or include the same or different material as that of the second mask layer **47**B.

[0046] A spacer layer **41** is formed over sidewalls of the mask layer **47** and the dummy gate layer **45**. The spacer layer **41** is or includes an insulating material, such as SiOCN, SiOC, SiCN or the like (or any of the materials described with reference to FIG. **1**) and may have a single-layer structure or a multi-layer structure including a plurality of dielectric layers, in accordance with some embodiments. The spacer layer **41** may be formed by depositing a spacer material layer (not shown) over the mask layer **47** and the dummy gate layer **45**. Portions of the spacer material layer between dummy gate structures **40** are removed using an anisotropic etching process, in accordance with some embodiments. In some embodiments, as shown in detail in FIGS. **4**B, **4**C, the spacer layer **41** includes a first spacer layer **41**A in contact with the nanostructure **22**C, the gate dielectric layer **43**, the dummy gate layer **45** and the first and second mask layers **47**A, **47**B. A second spacer layer **41**B of the spacer layer **41** may be in contact with the first spacer layer **41**A. The first spacer layer **41**A may be or include the same or different material as that of the second spacer layer **41**B.

[0047] In FIGS. **5**A and **5**B, source/drain openings **59** are formed by performing an etching process to etch the portions of protruding fins **32** and/or nanostructures **22**, **24** that are not covered by dummy gate structures **40**. The recessing may be anisotropic, such that the portions of fins **32** directly underlying dummy gate structures **40** and the spacer layer **41** are protected and are not substantially etched. The top surfaces of the recessed fins **32** may be substantially coplanar with the top surfaces of the isolation regions **36**, in accordance with some embodiments. The top surfaces of the recessed fins **32** may be lower than the top surfaces of the isolation regions **36**, in accordance with some other embodiments, as depicted in FIG. **5**B. FIG. **5**A depicts three vertical stacks of nanostructures **22**, **24** following the etching process for simplicity. In general, the etching process may be used to form fewer or additional vertical stacks of nanostructures **22**, **24** over the fins **32** than those depicted. In some embodiments, the second mask layer **47**B is exposed following the etching process, for example, due to removal of upper portions of the spacer layers **41**A, **41**B during the etching process. FIG. **5**B depicts fin spacers **41**F which are portions of the first and/or second spacer layers **41**A, **41**B that overlie the isolation regions **36** adjacent to respective fins **32**.

[0048] FIGS. **6**A-**7**B depict formation of inner spacers **74** in accordance with various embodiments.

[0049] In FIGS. **6**A, **6**B, a selective etching process is performed to recess end portions of the nanostructures **24** exposed by openings in the spacer layer **41** without substantially attacking the nanostructures **22**. After the selective etching process, recesses are formed in the nanostructures **24** at locations where the removed end portions used to be. Then, following formation of the recesses **64**, an inner spacer layer **74**L is formed to fill (partially or entirely) the recesses in the nanostructures **22** formed by the previous selective etching process. The inner spacer layer **74**L may be a suitable dielectric material, such as silicon carbon nitride (SiCN), silicon oxycarbonitride (SiOCN), or the like, formed by a suitable deposition method such as PVD, CVD, ALD, or the like.

[0050] In FIGS. **7**A and **7**B, following formation of the inner spacer layer **74**L, an etching process, such as an anisotropic etching process, is performed to remove portions of the inner spacer layer **74** disposed outside the recesses, for example, on sidewalls of the nanostructures **22** and the fins **32**. The remaining portions of the inner spacer layer **74**L (e.g., portions disposed inside the recesses in the nanostructures **24**) form the inner spacers **74**.

[0051] In FIG. **8**A, a first semiconductor layer **110**A is formed in the source/drain openings **59**. The first semiconductor layer **110**A is an undoped silicon layer in some embodiments that may be deposited or epitaxially grown on exposed surfaces of the fin **32**. The deposition may include one or more operations, such as a CVD, which may be an Ultra-High Vacuum Chemical Vapor Deposition (UHV-CVD), which allows for improved control of deposition rate and purity of the first semiconductor layer **110**A. In some embodiments, precursor gases containing silicon may be introduced into a processing chamber, and a reaction therebetween forms the silicon material, which deposits into the source/drain openings **59**. In some embodiments, the first semiconductor layer **110**A has an upper surface that is at a level substantially coplanar with an upper surface of the fin **32**.

[0052] FIG. **8**A also depicts formation of bottom insulators **800** and source/drain regions **82**N, **82**P or "source/drains **82**N, **82**P" in accordance with various embodiments. FIG. **8**A depicts formation of p-type source/drains **82**P and n-type source/drains **82**N, which may be different from each other in some respects. Namely, devices including n-type source/drains **82**N may benefit from inclusion of the bottom insulator **800**, whereas devices including p-type source/drains **82**P may benefit from exclusion of the bottom insulator **800**.

[0053] In FIG. **8**A, source/drain regions **82**N, **82**P are formed. In the illustrated embodiment, the source/drain regions **82**P are epitaxially grown from epitaxial material(s). In some embodiments, the source/drain regions **82**P exert stress in the respective channels **22**A, **22**B, **22**C, thereby improving performance. The source/drain regions **82**P are formed such that each dummy gate structure **40** is disposed between respective neighboring pairs of the source/drain regions **82**P. In some embodiments, the spacer layer **41** separates the source/drain regions **82**P from the dummy gate layer **45** by an appropriate lateral distance to prevent electrical bridging to subsequently formed gates of the resulting device. The source/drain regions **82**P may be or include Si:B, Si:Ga, SiGe:B, SiGe:B:Ga, SiGe:Sn, SiGe:B:Sn or the like. The source/drain regions **82**P may exert a compressive strain in the channel regions. The source/drain regions **82**N, **82**P may have surfaces raised from respective surfaces of the first semiconductor layer **110**A and may have facets. Neighboring source/drain regions **82**P may merge in some embodiments to form a singular source/drain region **82**P adjacent two neighboring fins **32**.

[0054] In the illustrated embodiment, following formation of the source/drain regions **82**P, the optional bottom insulator **800** may be formed. Formation of the bottom insulator **800** may include a suitable deposition operation, such as an LPCVD, PECVD, ALD, or the like. The bottom insulator **800** may be or include SiN or another suitable dielectric material. Thickness of the bottom insulator **800** may be in a range of about 2 nm to about 4 nm. During formation of the bottom insulator **800**, a dielectric layer **820** may be formed on the upper surface of the source/drain region **82**P. The dielectric layer **820** may have the same composition (e.g., SiN) and thickness as those of the bottom insulator **800**. The bottom insulator **800** may be present on the first semiconductor layer **110**A, such as in direct contact with an upper surface of the first semiconductor layer **110**A. Side surfaces of the bottom insulator **800** may be in direct contact with side surfaces of a bottommost inner spacer of the inner spacers **74**.

[0055] Following formation of the bottom insulator **800**, the source/drain regions **82**N are epitaxially grown from epitaxial material(s). Due to the bottom insulator **800**, the source/drain regions **82**N may grow from the channels **22** without growing from the first semiconductor layer **110**A. In some embodiments, the source/drain regions **82**N exert stress in the respective channels **22**A, **22**B, **22**C, thereby improving performance. The source/drain regions **82**N are formed such that each dummy gate structure **40** is disposed between respective neighboring pairs of the source/drain regions **82**N. In some embodiments, the spacer layer **41** separates the source/drain regions **82**N from the dummy gate layer **45** by an appropriate lateral distance to prevent electrical bridging to subsequently formed gates of the resulting device. The source/drain regions **82**N may be or include SiP, SiAs, SiSb, SiPAs, SiP:As:Sb or the like. The source/drain regions **82**N may exert a tensile strain in the channel regions due to presence of the bottom insulator **800**. The source/drain regions **82**N may have surfaces raised from respective surfaces of the fins **32** and may have facets. Neighboring source/drain regions **82**N may merge in some embodiments to form a singular source/drain region **82**N adjacent two neighboring fins **32**.

[0056] In some embodiments, the bottom insulator **800** and dielectric layer **820** are not formed. In such embodiments, the source/drain regions **82**N, **82**P grow from the first semiconductor layer **110**A and the nanosheets **22** and are in direct contact with the first semiconductor layer **110**A.

[0057] In FIG. **8**B, following formation of the source/drain regions **82**N, **82**P, the ILD **130** may be formed covering the source/drain regions **82**N, **82**P and abutting the spacer layer **41**. In some embodiments, the ESL **131** is formed prior to forming the ILD **130**. The ESL **131** may be formed by depositing a conformal thin layer of a dielectric material different from that of the ILD **130**, such as one or more of SiN, SiCN, SiC, SiOC, SiOCN, HfO2, ZrO2, ZrAlOx, HfAlOx, HfSiOx, Al2O3, or other suitable material. Following deposition of the ESL **131**, the ILD **130** may be deposited by a suitable process, such as a blanket deposition process, including PVD, CVD, ALD, or the like. The material of the ILD **130** may include silicon dioxide or a low-k dielectric material (e.g., a material having a dielectric constant (k-value) lower than the k-value (about 3.9) of silicon dioxide). The low-k dielectric material may include silicon oxynitride, phosphosilicate glass (PSG), borosilicate glass (BSG), borophosphosilicate glass (BPSG), undoped silicate glass (USG), fluorinated silicate glass (FSG), silicon oxycarbide (SiOxCy), Spin-On-Glass (SOG) or a combination thereof. The ILD **130** may be deposited by spin-on coating, CVD, Flowable CVD (FCVD), PECVD, PVD, or another deposition process.

[0058] In FIGS. **9**A-**10**C, following formation of the source/drains **82**N, **82**P, the ESL **131** and the ILD **130**, active gate structures **200** may be formed. A planarization process, such as a chemical mechanical polishing (CMP) process, is performed on the ILD **130** and the ESL **131**. The hard masks **47**A, **47**B and portions of the gate spacers **41** are also removed in the planarization process. After the planarization process, the dummy gate layers **45** are exposed. The top surfaces of the ILD **130** and the ESL **131** may be coplanar with the top surfaces of the dummy gate layers **45** and the gate spacers **41**.

[0059] Next, as depicted in FIGS. **9**A-**9**C, the dummy gate layer **45** is removed in an etching process, so that openings **92** are formed, corresponding to act **1400** of FIG. **14**. In some embodiments, the dummy gate layer **45** is removed by an anisotropic dry etch process. For example, the etching process may include a dry etch process using reaction gas(es) that selectively etch the dummy gate layer **45** without etching the spacer layer **41**. The dummy gate dielectric **43**, when present, may be used as an etch stop layer when the dummy gate layer **45** is etched. The dummy gate dielectric **43** may then be removed after the removal of the dummy gate layer **45**.

[0060] The nanostructures **24** are removed to release the nanostructures **22**. After the nanostructures **24** are removed, the nanostructures **22** form a plurality of nanosheets that extend horizontally (e.g., parallel to a major upper surface of the substrate **110**). In some embodiments, the nanostructures **24** are removed by a selective etching process using an etchant that is selective to the material of the nanostructures **24**, such that the nanostructures **24** are removed without substantially attacking the nanostructures **22**. In some embodiments, the etching process is an isotropic etching process using an etching gas, and optionally, a carrier gas, where the etching gas comprises F2 and HF, and the carrier gas may be an inert gas such as Ar, He, N2, combinations thereof, or the like.

[0061] In some embodiments, the nanostructures **24** are removed and the nanostructures **22** are patterned to form channel regions of both PFETs and NFETs. However, in some embodiments the nanostructures **24** may be removed and the nanostructures **22** may be patterned to form channel regions of NFETs, and nanostructures **22** may be removed and the nanostructures **24** may be patterned to form channel regions of PFETs. In some embodiments, the nanostructures **22** may be removed and the nanostructures **24** may be patterned to form channel regions of NFETs, and the nanostructures **24** may be removed and the nanostructures **22** may be patterned to form channel regions of PFETs. In some embodiments, the nanostructures **22** may be removed and the nanostructures **24** may be patterned to form channel regions of both PFETs and NFETs.

[0062] In some embodiments, the nanosheets **22** are reshaped (e.g. thinned) by a further etching process to improve gate fill window. The reshaping may be performed by an isotropic etching process selective to the nanosheets **22**. After reshaping, the nanosheets **22** may exhibit the dog bone shape in which middle portions of the nanosheets **22** are thinner than peripheral portions of the nanosheets **22** along the X direction.

[0063] Then, replacement gates **200** are formed. The replacement gates **200** may be referred to as active gates **200** or gate structures **200**. The gate structures **200** may be formed by a series of deposition operations, such as ALD cycles, that deposit the various layers of the gate structure **200** in the openings, described below with reference to FIG. **11**. Formation of the gate structure **200** is described in detail with reference to FIG. **11** to provide context for understanding the embodiments described with reference to FIGS. **12**A-**12**Q.

[0064] FIG. **11** is a detailed view of a portion of the gate structure **200**. The gate structure **200** generally includes the interfacial layer (IL, or "first IL" below) **210**, at least one gate dielectric layer **600**, the work function metal layer **900**, and the gate fill layer **290**. In some embodiments, each replacement gate **200** further includes at least one of a second interfacial layer **240** or a second work function layer **700**.

[0065] With reference to FIG. **11**, in some embodiments, the first IL **210** includes an oxide of the semiconductor material of the substrate **110**, e.g. silicon oxide. In other embodiments, the first IL **210** may include another suitable type of dielectric material. The first IL **210** has a thickness in a range between about 5 angstroms and about 50 angstroms.

[0066] Still referring to FIG. **11**, the gate dielectric layer **600** is formed over the first IL **210**. In some embodiments, an atomic layer deposition (ALD) process is used to form the gate dielectric layer **600** to control thickness of the deposited gate dielectric layer **600** with precision. In some embodiments, the ALD process is performed using between about 40 and 80 deposition cycles, at a temperature range between about 200 degrees Celsius and about 300 degrees Celsius. In some embodiments, the ALD process uses HfCl4 and/or H20 as precursors. Such an ALD process may form the first gate dielectric layer **220** to have a thickness in a range between about 10 angstroms and about 100 angstroms.

[0067] In some embodiments, the gate dielectric layer **600** includes a high-k dielectric material, which may refer to dielectric materials having a high dielectric constant that is greater than a dielectric constant of silicon oxide (k≈3.9). Exemplary high-k dielectric materials include HfO.sub.2, HfSiO, HfSiON, HfTaO, HfTiO, HfZrO, ZrO.sub.2, Ta.sub.2O.sub.5, or combinations thereof. In other embodiments, the gate dielectric layer **600** may include a non-high-k dielectric material such as silicon oxide. In some embodiments, the gate dielectric layer **600** includes more than one high-k dielectric layer, of which at least one includes dopants, such as lanthanum, magnesium, yttrium, or the like, which may be driven in by an annealing process to modify threshold voltage of the nanostructure devices **20**A, **20**B.

[0068] With further reference to FIG. **11**, an optional second IL **240** is formed on the gate dielectric layer **600**, and the second work function layer **700** is formed on the second IL **240**. The second IL **240** promotes better metal gate adhesion on the gate dielectric layer **600**. In many embodiments, the second IL **240** further provides improved thermal stability for the gate structure **200** and serves to limit diffusion of metallic impurities from the work function metal layer **900** and/or the work function barrier layer **700** into the gate dielectric layer **600**. In some embodiments, formation of the second IL **240** is accomplished by first depositing a high-k capping layer (not illustrated for simplicity) on the gate dielectric layer **600**. The high-k capping layer comprises one or more of the following: HfSiON, HfTaO, HfTiO, HfTaO, HfAlON, HfZrO, or other suitable materials, in various embodiments. In one embodiment, the high-k capping layer comprises titanium silicon nitride (TiSiN). In some embodiments, the high-k capping layer is deposited by an ALD using about 40 to about 100 cycles at a temperature of about 400 degrees C. to about 450 degrees C. A thermal anneal is then performed to form the second IL **240**, which may be or comprise TiSiNO, in some embodiments. Following formation of the second IL **240** by thermal anneal, an atomic layer etch (ALE) with artificial intelligence (AI) control may be performed in cycles to remove the high-k capping layer while substantially not removing the second IL **240**. Each cycle may include a first pulse of WCl.sub.5, followed by an Ar purge, followed by a second pulse of O.sub.2, followed by another Ar purge. The high-k capping layer is removed to increase gate fill window for further multiple threshold voltage tuning by metal gate patterning.

[0069] Further in FIG. **11**, after forming the second IL **240** and removing the high-k capping layer, the work function barrier layer **700** is optionally formed on the gate structure **200**, in accordance with some embodiments. The work function barrier layer **700** is or comprises a metal nitride, such as TiN, WN, MoN, TaN, or the like. In a specific embodiment, the work function barrier layer **700** is TiN. The work function barrier layer **700** may have thickness ranging from about 5 A to about 20 A. Inclusion of the work function barrier layer **700** provides additional threshold voltage tuning flexibility. In general, the work function barrier layer **700** increases the threshold voltage for NFET transistor devices, and decreases the threshold voltage (magnitude) for PFET transistor devices.

[0070] The work function metal layer **900**, which may include at least one of an N-type work function metal layer, an in-situ capping layer, or an oxygen blocking layer, is formed on the work function barrier layer **700**, in some embodiments. The N-type work function metal layer is or comprises an N-type metal material, such as TiAlC, TiAl, TaAlC, TaAl, or the like. The N-type work function metal layer may be formed by one or more deposition methods, such as CVD, PVD, ALD, plating, and/or other suitable methods, and has a thickness between about 10 A and 20 A. The in-situ capping layer is formed on the N-type work function metal layer. In some embodiments, the in-situ capping layer is or comprises TiN, TiSiN, TaN, or another suitable material, and has a thickness between about 10 A and 20 A. The oxygen blocking layer is formed on the in-situ capping layer to prevent oxygen diffusion into the N-type work function metal layer, which would cause an undesirable shift in the threshold voltage. The oxygen blocking layer is formed of a dielectric material that can stop oxygen from penetrating to the N-type work function metal layer, and may protect the N-type work function metal layer from further oxidation. The oxygen blocking layer may include an oxide of silicon, germanium, SiGe, or another suitable material. In some embodiments, the oxygen blocking layer is formed using ALD and has a thickness between about 10 A and about 20 A.

[0071] FIG. **11** further illustrates the metal core layer **290**. In some embodiments, a glue layer (not separately illustrated) is formed between the oxygen blocking layer of the work function metal layer and the metal core layer **290**. The glue layer may promote and/or enhance the adhesion between the metal core layer **290** and the work function metal layer **900**. In some embodiments, the glue layer may be formed of a metal nitride, such as TiN, TaN, MoN, WN, or another suitable material, using ALD. In some embodiments, thickness of the glue layer is between about 10 A and about 25 A. The metal core layer **290** may be formed on the glue layer, and may include a conductive material such as tungsten, cobalt, ruthenium, iridium, molybdenum, copper, aluminum, or combinations thereof. In some embodiments, the metal core layer **290** may be deposited using methods such as CVD, PVD, plating, and/or other suitable processes. In some embodiments, a seam **510**, which may be an air gap, is formed in the metal core layer **290** vertically between the channels **22**A, **22**B, **22**C. In some embodiments, the metal core layer **290** is conformally deposited on the work function metal layer **900**. The seam **510** may form due to sidewall deposited film merging during the conformal deposition. In some embodiments, the seam **510** is not present between the neighboring channels **22**A, **22**B, **22**C.

[0072] In some embodiments, one or more of metal layers including the core layer **290** and the work function layers **700**, **900** in PFET devices can include Ti, Al, Zn, W, Nb, Co and the like and total thickness of the combination of metal layers on the gate dielectric layer **600** may be in a range of about 0.5 nm to about 20 nm. In some embodiments, one or more of metal layers including the core layer **290** and the work function layers **700**, **900** in NFET devices can include Ti and/or Al and the like and total thickness of the combination of metal layers on the gate dielectric layer **600** may be in a range of about 0.5 nm to about 20 nm.

[0073] FIGS. **12**A-**12**Q are cross-sectional side views of formation of a portion of the gate structure **200** of the IC device **10** in accordance with various embodiments. The dielectric layer **600**, such as HfO2, SiO2, and ZrO2, is beneficial for improving turning on of a field effect transistor, such as the nanostructure transistor **20**A. Quality of the dielectric layer **600**, including thickness, vacancies, heterogeneous atom contamination and the like can degrade threshold voltage (Vt), channel resistance (Rch), ring performance (RO %), time-dependent-dielectric-breakdown (TDDB) and maximum voltage degradation, especially after long periods of operation, e.g., about ten years. In devices that include nanosheets such as described with reference to FIGS. **1**A-**11**, dielectrics at corners of nanosheets **22** undergo dry bombardment and may experience a stress concentration effect that degrades film quality. Then, a subsequent wet chemical etch or clean for metal removal may damage the dielectric layer **600**.

[0074] In the embodiments described with reference to FIGS. **11**A-**11**Q, a high-selectivity wet-etching process simultaneously etches metal (e.g., the work function layers **700**, **900**) without substantially removing the dielectric layer **600**.

[0075] In FIGS. **12**A, **12**B, following formation of the IL **210** and the high-k dielectric layer **600**, a capping layer **710** may be formed on and wrapping around the channels **22**A, **22**B, **22**C and the fins **32**, corresponding to act **1400** of FIG. **14**. The capping layer **710** may wrap around the channels **22** on all sides in the cross-sectional profile along the YZ plane, as depicted in FIG. **12**A and may partially wrap around the fins **32**. The capping layer **710** may be or include a dielectric material, such as SiN, SiCN, SiC, SiOC, SiOCN, HfO2, ZrO2, ZrAlOx, HfAlOx, HfSiOx, Al.sub.2O.sub.3 or the like. In some embodiments, the capping layer **710** is an Al.sub.2O.sub.3 layer. The capping layer **710** may be deposited by a suitable deposition process, such as a PVD, CVD, ALD or the like. Thickness of the capping layer **710** may be in a range of about 1 nm to about 5 nm, such as about 3 nm. Following the deposition, a first gap D**1** may be present between adjacent side surfaces of the capping layer **710**. The first gap D**1** may be in a range of about 1 nm to about 6 nm, such as about 3.5 nm. A later process deposits a thin layer of a transition metal nitride, such as TiN, TaN, onto the channels **22**. Forming the capping layer **710** is beneficial to prevent the transition metal nitride layer from filling gaps between vertically adjacent channels **22**, such as the channel **22**A and the channel **22**B.

[0076] In FIGS. **12**C, **12**D, following formation of the capping layer **710**, a thinned capping layer **710**′ is formed by thinning the capping layer **710**. Thinning the capping layer **710** to form the thinned capping layer **710**′ can be beneficial to enlarge poly spacing, such as from about 3.5 nm (e.g., first gap D**1**) to about 6.5 nm or more (e.g., second gap D**2**). In some embodiments, the second gap D**2** is in a range of about 3 nm to about 10 nm, such as about 6 nm. The second gap D**2** is larger than the first gap D**1**. Thinning the capping layer **710** may include one or more etching operations, which may be isotropic wet etch operations. The thinned capping layer **710**′ may have thickness that is in a range of about 1 nm to about 2 nm, such as about 1.7 nm, which is about or slightly more than half the thickness of the capping layer **710**. In some embodiments, thinning of the capping layer **710** to form the thinned capping layer **710**′ is not performed.

[0077] In FIGS. **12**E, **12**F, following optional thinning of the capping layer **710** to form the thinned capping layer **710**′, a patterned mask **400** is formed that protects a first region **610** of the device **10** while exposing a second region **620** of the device **10**. In some embodiments, the patterned mask **400** is or includes a bottom antireflective coating (BARC) layer **400**. Prior to forming the BARC layer **400**, the surface of the device **10** may be cleaned to remove particles or unwanted materials that can interfere with the BARC deposition. Then, the BARC layer **400** may be deposited over the entire surface of the device **10**, covering the stacks of nanosheets **22** and gaps in between. The

BARC layer **400** may be deposited via spin coating or chemical vapor deposition (CVD). After deposition, the BARC layer **400** may be baked to remove solvent and improve properties thereof. Then, a photoresist layer may be applied on top of the BARC layer **400**. A pattern is then transferred onto the photoresist layer via a photolithography process that involves exposing the photoresist layer to light reflected from a mask that has the selected pattern. The exposed photoresist layer is developed, washing away either the exposed or unexposed areas (depending on whether a positive or negative photoresist is used), exposing the underlying BARC layer **400** in a pattern that matches the mask. Then, the exposed BARC areas are etched away, for example by a plasma etch process. The etching selectively removes the BARC where the photoresist has been washed away, transferring the pattern from the photoresist to the BARC layer **400**. After etching, remaining photoresist is stripped away, leaving behind a patterned BARC layer **400** that corresponds to the selected design.

[0078] In FIGS. **12**E, **12**F, portions of the thinned capping layer **710**′ (or the capping layer **710** when not thinned) that are exposed by the BARC layer **400** are removed via one or more suitable etching operations. Removal of the thinned capping layer **710**′ may be in the second region **620**, which may be a PFET region **610** of the device **10**. The removal may be via an etching operation, such as an isotropic wet etch, that is selective to material of the thinned capping layer **710**′ without substantially attacking the gate dielectric layer **600** and/or the substrate **110**. The removal may completely remove the thinned capping layer **710** from surfaces of the nanosheets **22** in the second region **620**, such that the gate dielectric layer **600** is completely exposed in the second region **620**.

[0079] In FIGS. **12**G, **12**H, following removal of the thinned capping layer **710**′ or the capping layer **710** in the second region **620**, the thinned capping layer **710**′ or the capping layer **710** is thinned to remove portions thereof that are on an upper surface of the topmost nanosheet **22**C and on side surfaces or sidewalls of the nanosheets **22**A, **22**B, **22**C and the fin **32**. This results in the dummy plugs **710**″ that are positioned between adjacent nanosheets **22** (e.g., the nanosheets **22**A, **22**B) and between the bottommost nanosheet **22**A and the fin **32**. The dummy plugs **710**″ may have width in the Y-axis direction that is less than width of the nanosheets **22**A, **22**B, **22**C in the Y-axis direction. Each of the dummy plugs **710**″ may extend from the gate dielectric layer **600** on one of the nanosheets **22** to the gate dielectric layer **600** on another of the nanosheets **22** or the fin **32** adjacent to the one. The dummy plugs **710**″ may be in direct contact with the gate dielectric layer **600** and may have side surfaces that are exposed. Following formation of the dummy plugs **710**″ by removing the portions described, a third gap D**3** is present between adjacent sidewalls, as depicted in FIG. **12**H. The third gap D**3** may be in a range of about 5 nm to about 15 nm, such as about 10 nm. The third gap D**3** exceeds the second gap D**2** and the first gap D**1**, which further increases poly spacing.

[0080] In some embodiments, removal of the capping layer **710** and/or the thinned capping layer **710**′ may result in some slight removal of the underlying dielectric layer **600**. As such, after the etching operation described with reference to FIGS. **12**G, **12**H, exposed regions of the gate dielectric layer **600** in the first region **610** may have thickness that is slightly less than protected regions of the gate dielectric layer **600** in the first regions **610** (e.g., regions in direct contact with the dummy plugs **710**″) and that slightly exceeds thickness of the gate dielectric layer **600** in the second region **620**. In some embodiments, the thickness of the gate dielectric layer **600** in the second region **620** (e.g., around the nanosheets **22**) may be substantially uniform whereas that of the gate dielectric layer **600** in the first region **620** may be non-uniform.

[0081] In FIGS. **12**I, **12**J, following formation of the dummy plugs **710**″, a transition metal nitride layer **720** (or “layer **720**”), such as a TiN layer, is formed on exposed surfaces of the gate dielectric layer **600**, the dummy plugs **710**″ and the substrate **110**, corresponding to act **1500** of FIG. **14**. The layer **720** may be deposited on the surfaces by a suitable deposition operation, which may be a PVD, CVD, ALD or the like. Thickness of the layer **720** may be in a range of about 1 nm to about 5 nm, such as about 3 nm. Following deposition of the layer **720**, a fourth gap D**4** may be present that is smaller than the third gap D**3** and is in a range of about 2 nm to about 6 nm, such as about 4 nm. Due to the dummy plugs **710**″, the transition metal nitride layer **720** in the first region **610** may not wrap completely around the nanosheets **22** and fill the gaps therebetween. In the second region **620**, the transition metal nitride layer **720** may wrap completely around the nanosheets **22** and fill the gaps therebetween. The layer **720** will be removed in the first region **610** in a later operation. Having the dummy plugs **710**″ in the gaps between the nanosheets **22** instead of the layer **720** makes it easier to clear the gaps of material prior to forming layers of the gate structure **200**. Namely, it may be easier to remove the dummy plugs **710**″ from between adjacent nanostructures **22** than it is to remove the transition metal nitride layer **720** from between the adjacent nanostructures **22**.

[0082] In FIGS. **12**K, **12**L, following formation of the transition metal nitride layer **720**, a second patterned mask layer **410** is formed that covers the second region **620** while exposing the first region **610**, corresponding to act **1600** of FIG. **14**. As such, the layer **720** is exposed in the first region **610**. In some embodiments, the second patterned mask layer **410** is or includes a BARC layer. Then, the layer **720** is removed in the first region **610** leaving a patterned layer **720**′ in the second region **620** and exposing portions of the gate dielectric layer **610** outside the dummy plugs **710**″. The layer **720** may be removed by a suitable etching operation, such as a wet etch. Following removal of the layer **720**, remaining portions of the gate dielectric layer **600** in the first region **610** are exposed by removing the dummy plugs **710**″, such that the gate dielectric layer **600** is fully exposed in the first region **610**. As discussed with reference to FIGS. **12**G and **12**H, due to differences in number and sequence of etching operations performed on the gate dielectric layer **600** in the first region **610**, as well as presence of the dummy plugs **710**″, thickness of the gate dielectric layer **600** in the first region **610** may be non-uniform. For example, thickness of portions of the gate dielectric layer **600** that were in direct contact with the dummy plugs **710**″ may slightly exceed thickness of other portions of the gate dielectric layer **600** that were exposed by the dummy plugs **710**″.

[0083] FIGS. **12**M, **12**N, **12**O and **12**P are cross-sectional views of a single stack of nanostructures **22**A, **22**B, **22**C of the first region **610** following a removal process that removes the layer **720** and the dummy plugs **710**″ as described with reference to FIGS. **12**K, **12**L and FIG. **15**.

[0084] In FIG. **12**M, the layer **720** is exposed. For example, the layer **720** may be positioned in the first region **610** while the second region **620** is protected by the second patterned mask **410**. When the transition metal nitride layer **720** is deposited on the gate dielectric layer **600**, some intermixing between the layers **720**, **600** occurs, which may be referred to as natural intermixing.

[0085] Then, in FIG. **12**N, a plasma etching or plasma bombarding is performed to form the second patterned mask layer **410**, corresponding to act **2100** of FIG. **15**. Namely, a BARC layer may be formed on the first and second regions **610**, **620**. Then a patterned photoresist layer may be formed on the BARC layer. Exposed portions of the BARC layer, such as in the first region, are then removed via a plasma etch process, which may include plasma bombardment, which is a process where a suitable gas is introduced and ionized in a vacuum chamber, creating a plasma. The ionization occurs when energy, usually from radio-frequency (RF) power, is applied, leading to the formation of positively charged ions and free electrons. The ions are then accelerated by an electric field towards the BARC layer, where they energetically collide with the surface thereof. These collisions can physically knock off atoms (physical sputtering), react chemically to form volatile byproducts, or cause radiation damage to the material. The intensity and effects of the bombardment are influenced by factors such as the power of the RF source, chamber pressure, and etching duration. The process may be selective and anisotropic, removing the BARC efficiently while reducing damage to underlying or adjacent materials, which can be beneficial for maintaining precise feature dimensions.

[0086] The inventors have realized that, although plasma bombardment can reduce damage to underlying or adjacent materials, such as the gate dielectric layer **600**, it is not sufficient in all respects. Namely, as depicted in FIG. **12**N, in corner regions **600**X of the gate dielectric layer **600** wrapped around at least the uppermost channel **22**C, deep intermixing between the gate dielectric layer **600** and the transition metal nitride layer **720** may occur. In some instances, the gate dielectric layer **600** wrapped around the channel **22**B immediately below the uppermost channel **22**C may also have deep intermixing in corner regions thereof.

[0087] In FIG. **12**O, one or more wet etch operations are performed that remove the layer **720** and the dummy plugs **710**″, resulting in the structure shown and corresponding to act **1700** of FIG. **14**. In the corner regions **600**X, voids **600**X′ may be formed by removal of material of the gate dielectric layer **600** during the wet etch operation(s). The voids **600**X′ can result in degradation of threshold voltage Vt, channel resistance Rch, ring performance RO %, time-dependent-dielectric-breakdown (TDDB), maximum voltage and the like. The inventors have realized that performing one or more etches that use a basic or first etchant, such as an etchant including NH4OH:H2O2:H2O in a ratio of about 1:5:25 can result in formation of the voids **600**X′. In some embodiments, instead of using the basic etchant, a slightly acidic or mildly acidic or second etchant is used to remove the layer **720**, which results in little to no removal of the gate dielectric layer **600** in the corner regions **600**X, thereby reducing the formation of voids **600**X′ and improving one or more of the performance metrics just listed. In some embodiments, a third etchant, such as H2O2:H2O in a ratio of about 1:5 may be included in the one or more wet etch operations.

[0088] In some embodiments, a first etch operation is performed, followed by a second etch operation, followed by an optional third etch operation.

[0089] Following removal of the BARC **410** and exposure of the first region **610**, the first etch operation may be performed, corresponding to act **2200** of FIG. **15**. The first etch operation may be a wet etch operation that includes the second etchant. In some embodiments, the second etchant includes HCl:H2O2:H2O in a ratio of about 1:10:50. In some embodiments, the second etchant includes HCl in a concentration in a range of about 0.1 wt % to about 50 wt %. In some embodiments, the second etchant includes an organic acid in a concentration in a range of about 0.1 wt % to about 50 wt %. In some embodiments, the organic acid includes one or more of citric acid, formic acid, oxalic acid, or the like. The first etch operation may remove most of the layer **720**, such as at least about 90% of the layer **720**, at least about 95% of the layer **720**, at least about 99% of the layer **720** or another suitable amount of the layer **720**.

[0090] The second etch operation follows the first etch operation, corresponding to act **2300** of FIG. **15**, and may include an oxidizing mixture as the third etchant, such as H2O2:H2O in a ratio of about 1:5. In some embodiments, a concentration of an oxidant in the mixture may be in a range of about 0.1 to about 10.sup.7 ppm. The oxidant may include H2O2, O3, or the like.

[0091] The optional third etch operation follows the second etch operation, corresponding to act **2400** of FIG. **15**, and may include the second etchant. Concentration of acid (e.g., HCl or organic acid) in the second etchant in the third etch operation may be in a range of about 0.1 wt % to about 50 wt %. In some embodiments, the concentration of acid is the same in the first etch operation and the third etch operation. In some embodiments, the concentration of acid is different in the third etch operation than in the first etch operation. For example, the concentration may be lower in the third etch operation than in the first etch operation or higher in the third etch operation than in the first etch operation. The third etch operation may be performed for a different length of time than that of the first etch operation. For example, the third etch operation may be performed for a length of time is shorter than that of the first etch operation.

[0092] FIG. **12**Q is a cross-sectional view depicting a channel **22** having a gate dielectric layer **600** thereon. The channel **22** has a middle region **810** in which upper and lower surfaces of the channel **22** are substantially horizontal and flat. The channel **22** has left and right end regions **830**L, **830**R that are on either side of the middle region **810** and have curved side surfaces. In some embodiments, length of the channel **22** is in a range of about 8 nm to about 70 nm. The channel **22** has a first diagonal or "sheet diagonal" L**1** that is in a range of about 10 nm to about 65 nm. The combination of the channel **22** and the gate dielectric layer **600** has a second diagonal L**2**. The second diagonal L**2** exceeds the first diagonal L**1**, such as 0.2 nm<(L**2**−L**1**)<4 nm. The first diagonal L**1** may be offset from a horizontal line L**3** of the channel **22** by a first angle θ**1** that is in a range of about −45° to about 45°. The second diagonal L**2** may be offset from the first diagonal by a second angle θ**2** that is in a range of about −45° to about 45°. In some embodiments, the first diagonal L**1**, the second diagonal L**2**, the first angle θ**1** and the second angle θ**2** are related by a relationship, such as 0.3 nm<(L**2**θ**2**−L**1**θ**1**)<6.3 nm.

[0093] The gate dielectric layer **600** includes corner regions or portions **600**C. In the corner regions **600**C, concentration of TiN in the gate dielectric layer **600** may exceed that outside of the corner regions **600**C. Namely, deep intermixing between the layer **720** and the gate dielectric layer **600** may occur during plasma bombardment of the BARC **410**, such that a relatively higher concentration of TiN is present in the corner regions **600**C. The corner regions **600**C generally are only on the upper side of the channel **22** and not on the lower side of the channel **22**. This is because the upper side of the channel **22** and gate dielectric layer **600** are in a direct path of the plasma bombardment, whereas the lower side of the channel **22** and gate dielectric layer **600** are protected somewhat due to being in an indirect path of the plasma bombardment.

[0094] In some embodiments, although a mildly acidic etchant is used in the first and optionally third etching operations, some slight removal of material in the corner regions **600**C due to the deep intermixing may still occur. As such, thickness of the gate dielectric layer **600** in the corner regions **600**C may be slightly less than outside of the corner regions **600**C. For example, a ratio of the thickness in the corner regions **600**C to the thickness outside the corner regions **600**C may be in a range of about 80% to about 99.5%, such as in a range of about 90% to about 99%. The thickness in the corner regions **600**C may be conserved by using the mildly acidic etchant described with reference to FIG. **12**P.

[0095] Because the BARC **410** over the first region **610** is removed via plasma bombardment whereas the BARC **410** over the second region **620** may be removed via ashing or another suitable method, the gate dielectric **600** in the second region **620** may not undergo deep intermixing with the layer **720**. As such, the corner regions **600**C in the first region **610** may be different in some respects than the corner regions **600**C in the second region **620**. For example, concentration of transition metal nitride in the corner regions **600**C in the first region **610** may exceed concentration of transition metal nitride in the corner regions **600**C in the second region **620**. In another example, some slight removal of the deeply intermixed material in the corner regions **600**C may take place during the removal of the layer **720** in the first region **610**, such that thickness of the gate dielectric **600** in the corner regions **600**C in the first region **610** may be less than that of the gate dielectric **600** in the corner regions **600**C in the second region **620**. For example, a ratio of the thickness in the corner regions **600**C in the first region **610** to the thickness in the corner regions **600**C in the second region **620** may be in a range of about 80% to about 99.5%, such as in a range of about 90% to about 99%.

[0096] In FIG. **13**, following formation of the gate structures **200**, source/drain openings may be formed in the ILD **130** and source/drain contacts **120** may be formed in the source/drain openings. The resulting structure is shown in FIGS. **15**A, **15**B. Silicide regions **118** and the source/drain contacts **120** are formed on the source/drain regions **82**N, **82**P.

[0097] In some embodiments, the silicide layers **118** are formed prior to formation of the source/drain contacts **120**. For example, an N-type or P-type metal layer may be formed as a conformal thin layer over exposed portions of the source/drain regions **82**N, **82**P. The metal layer may be or include one or more of Ni, Co, Mn, W, Fe, Rh, Pd, Ru, Pt, Ir, Os or the like. In some embodiments, the metal layer is or includes one or more of Ti, Cr, Ta, Mo, Zr, Hf, Sc, Ys, Ho, Tb, Gd, Lu, Dy, Er, Yb or another suitable material. Following formation of the metal layer, the silicide layers **118** may be formed by annealing the device **10**. Following the anneal, the silicide layers **118** may be or include one or more of NiSi, CoSi, MnSi, Wsi, FeSi, RhSi, PdSi, RuSi, PtSi, IrSi, OsSi, TiSi, CrSi, TaSi, MoSi, ZrSi, HfSi, ScSi, Ysi, HoSi, TbSI, GdSi, LuSi, DySi, ErSi, YbSi or the like. Silicide of the silicide layers **118** may diffuse into regions below the ESL **131**. Thickness of the silicide layers **118** may be in a range of about 1 nm to about 10 nm. Below about 1 nm, contact resistance may be too high. Above about 10 nm, the silicide layers **118** may short with the channels **22**B.

[0098] Following formation of the silicide layers **118**, the source/drain contacts **120** are formed by filling the openings over the source/drain regions **82**N, **82**P with, for example, a liner layer and a fill layer. In some embodiments, the source/drain contacts **120** are formed by depositing a material that is or includes a conductive material such as Co, W, Ru, combinations thereof, or the like. In some embodiments, the source/drain contacts **120** are or include a Co-, W- or Ru-based compound or alloy including one or more elements, such as Zr, Sn, Ag, Cu, Au, Al, Ca, Be, Mg, Rh, Na, Ir, W, Mo, Zn, Ni, K, Co, Cd, Ru, In, Os, Si, Ge, Mn, combinations thereof, or the like. The source/drain contacts **120** land on the silicide layer **118** and are in contact with the ESL **131**. Description of the device **10** and illustration thereof in many of the figures is given with reference to GAAFETs including vertical stacks of the nanostructures **22**. In some embodiments, the silicide layers **118** and the source/drain contacts **120** are formed in and on source/drain regions **82**N, **82**P of FinFET devices.

[0099] Additional processing may be performed to finish fabrication of the nanostructure devices **20**. For example, gate contacts (or gate vias) may be formed to electrically couple to the gate structures **200**. An interconnect structure may then be formed over the source/drain contacts **120** and the gate contacts. The interconnect structure may include a plurality of dielectric layers (including, for example, a second ILD) surrounding metallic features, including conductive traces and conductive vias, which form electrical connection between devices on the substrate **110**, such as the nanostructure devices **20**, as well as to IC devices external to the IC device **10**.

[0100] Embodiments may provide advantages. By etching the layer **720** using a mildly acidic etchant following plasma bombardment during BARC removal, etching of corner regions of the gate dielectric **600** that have deep intermixing with the transition metal nitride layer **720** is reduced or eliminated, which improves quality (e.g., thickness uniformity) of the gate dielectric **600**. This can reduce lifetime degradation of threshold voltage (Vt), channel resistance (Rch), ring performance (RO %), time-dependent-dielectric-breakdown (TDDB) and maximum voltage of devices including the gate dielectric **600**.

[0101] In accordance with at least one embodiment, a method includes: forming a stack of alternating first semiconductor channels and second semiconductor layers on a substrate; releasing the first semiconductor channels by removing the second semiconductor layers; forming a gate dielectric on the first semiconductor channels; forming a transition metal nitride layer on the gate dielectric; and exposing the gate dielectric in a first region of the substrate by removing the transition metal nitride layer. The removing includes: performing a first etch using an acidic etchant; and after the performing a first etch, performing a second etch using an oxidizing etchant.

[0102] In accordance with at least one embodiment, a method includes: releasing first semiconductor channels of a stack of alternating the first semiconductor channels and second semiconductor layers by removing the second semiconductor layers; forming a gate dielectric on the first semiconductor channels; forming a plurality of dummy plugs between adjacent pairs of the first semiconductor channels in a first region; forming a transition metal nitride layer on the gate dielectric in the first region and a second region; forming a bottom antireflective coating (BARC) layer on the transition metal nitride layer in the first and second regions; exposing the first region by patterning the BARC layer via plasma bombardment; and exposing the gate dielectric in the first region by removing the transition metal nitride layer. The removing includes: performing a first etch using a first acidic etchant; after the performing a first etch, performing a second etch using an oxidizing etchant; and after the performing a second etch, performing a third etch using a second acidic etchant.

[0103] In accordance with at least one embodiment, a device includes: a first stack of nanostructures in a first region; a second stack of nanostructures in a second region; and a first gate structure wrapping around the first stack of nanostructures. The first gate structure includes: a first gate dielectric having at least two first corner regions, the first corner regions being positioned at first upper corners of a first uppermost nanostructure of the first stack of nanostructures; and a first gate metal on the first gate dielectric. The device further includes a second gate structure wrapping around the second stack of nanostructures. The second gate structure includes: a second gate dielectric having at least two second corner regions, the second corner regions being positioned at second upper corners of a second uppermost nanostructure of the second stack of nanostructures; a transition metal nitride layer on the second gate dielectric; and a second gate metal on the transition metal nitride layer. Concentration of material of the transition metal nitride layer intermixed with the first gate dielectric exceeds that of the second gate dielectric.
[0104] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method, comprising: forming a stack of alternating first semiconductor channels and second semiconductor layers on a substrate; releasing the first semiconductor channels by removing the second semiconductor layers; forming a gate dielectric on the first semiconductor channels; forming a transition metal nitride layer on the gate dielectric; and exposing the gate dielectric in a first region of the substrate by removing the transition metal nitride layer, the removing including: performing a first etch using an acidic etchant; and after the performing a first etch, performing a second etch using an oxidizing etchant.
2. The method of claim 1, wherein the performing a first etch includes performing the first etch using the acidic etchant that includes HCl:H2O2:H2O in a ratio of about 1:10:50.
3. The method of claim 1, wherein the performing a first etch includes performing the first etch using the acidic etchant having HCl at a concentration in a range of about 0.1 wt % to about 50 wt %.
4. The method of claim 1, wherein the performing a first etch includes performing the first etch using the acidic etchant having an organic acid at a concentration in a range of about 0.1 wt % to about 50 wt %.
5. The method of claim 1, wherein the performing a second etch includes performing the second etch using the oxidizing agent having H2O2 in a range of about 0.1 ppm to about 10.sup.7 ppm.
6. The method of claim 1, wherein the performing a second etch includes performing the second etch using the oxidizing agent having O3 in a range of about 0.1 ppm to about 10.sup.7 ppm.
7. The method of claim 1, further comprising: forming a bottom antireflective coating (BARC) layer on the transition metal nitride layer in the first region; and prior to the removing the transition metal nitride layer, exposing the transition metal nitride layer in the first region by removing the BARC layer via plasma bombardment.
8. A method, comprising: releasing first semiconductor channels of a stack of alternating the first semiconductor channels and second semiconductor layers by removing the second semiconductor layers; forming a gate dielectric on the first semiconductor channels; forming a plurality of dummy plugs between adjacent pairs of the first semiconductor channels in a first region; forming a

transition metal nitride layer on the gate dielectric in the first region and a second region; forming a bottom antireflective coating (BARC) layer on the transition metal nitride layer in the first and second regions; exposing the first region by patterning the BARC layer via plasma bombardment; and exposing the gate dielectric in the first region by removing the transition metal nitride layer, the removing including: performing a first etch using a first acidic etchant; after the performing a first etch, performing a second etch using an oxidizing etchant; and after the performing a second etch, performing a third etch using a second acidic etchant.

9. The method of claim 8, wherein the performing a third etch includes using the second acidic etchant including a different acid than that of the first acidic etchant.

10. The method of claim 8, wherein the performing a third etch includes using the second acidic etchant including a same acid as that of the first acidic etchant at a different concentration than in that of the first acidic etchant.

11. The method of claim 10, wherein the performing a third etch includes using the second acidic etchant including a same acid as that of the first acidic etchant at a lower concentration than in that of the first acidic etchant.

12. The method of claim 8, wherein the performing a third etch includes performing the third etch for a time that is shorter than that of the first etch.

13. The method of claim 8, wherein the forming a plurality of dummy plugs includes forming a dummy plug that is in direct contact with the gate dielectric on a first channel of the first semiconductor channels and is in direct contact with the gate dielectric on a second channel of the first semiconductor channels.

14. The method of claim 8, further comprising: after the removing the transition metal nitride layer, removing the plurality of dummy plugs.

15. A device, comprising: a first stack of nanostructures in a first region; a second stack of nanostructures in a second region; a first gate structure wrapping around the first stack of nanostructures, the first gate structure including: a first gate dielectric having at least two first corner regions, the first corner regions being positioned at first upper corners of a first uppermost nanostructure of the first stack of nanostructures; and a first gate metal on the first gate dielectric; and a second gate structure wrapping around the second stack of nanostructures, the second gate structure including: a second gate dielectric having at least two second corner regions, the second corner regions being positioned at second upper corners of a second uppermost nanostructure of the second stack of nanostructures; a transition metal nitride layer on the second gate dielectric; and a second gate metal on the transition metal nitride layer; wherein concentration of material of the transition metal nitride layer intermixed with the first gate dielectric exceeds that of the second gate dielectric.

16. The device of claim 15, wherein first thickness of the first gate dielectric in the first corner regions is less than second thickness of the second gate dielectric in the second corner regions.

17. The device of claim 16, wherein a ratio of the first thickness to the second thickness is in a range of about 80% to about 99.5%.

18. The device of claim 16, wherein the ratio is in a range of about 90% to about 99%.

19. The device of claim 15, wherein first thickness of the first gate dielectric in the first corner regions is less than third thickness of the first gate dielectric outside the first corner regions.

20. The device of claim 15, wherein a ratio of the first thickness to the third thickness is in a range of about 80% to about 99.5%.